

displays the process name, process id, and user id when a new process is started in the system. The `proc:::exec-success` event occurs when a new process is started successfully.

The `-q` option suppresses output such as column headings and number of probes matched.

```
#!/usr/sbin/dtrace -qs
proc:::exec-success
{
    printf("%s(pid=%d) started by uid \
        - %d\n",execname, pid, uid);
}
```

5.4 DTrace for Web 2.0 Applications

The deployment stack for a typical Web 2.0 application is becoming increasingly complicated. The stack has JavaScript on the browser, multiple layers of API (for example, OpenSocial), an application server, a web server, a database, native code such as C/C++

or Java, and the operating system. A click on a web page can exercise multiple layers. Experts in one area might not be experts in another area, making performance tuning even more difficult. Each layer listed above has good debugging tools. Database administrators have GUI and command line tools to observe every detail of the database. Language debuggers such as gdb do a good job of providing visibility into both scripting and native languages in the stack. Tools such as JProbe, JMeter, and VisualVM provide visibility into Java code. Tools such as vmstat, iostat, mpstat, and truss provide visibility into the operating system.

These tools are important in performance tuning and understanding each layer in isolation. However, these tools do not provide insight into the entire system and interaction between the different layers. For example, these tools cannot help find the database queries that are executed when a user clicks a

JavaScript button on a browser. Also, some of these tools are not usable in production. For example, truss can slow down your application too much to be usable in production. Developers typically resort to creating custom debug code to address these issues. These pieces of code can be seen in debug enabled versions of applications. These debug versions of applications need special flags and reduce performance of the applications. Therefore, debug

Digit Subscription

Subscribe to the Digit developer newsletter at www.thinkdigit.com/newsletter